

Replication of Figure 4: Systematic and Idiosyncratic Variance Shares

Data Sources

Individual stock returns, market capitalization, and SIC codes are from the CRSP Daily Stock File via WRDS (1955–2025). Market excess returns and the risk-free rate are from Ken French’s daily Fama-French factors. CRSP is filtered to common equity (SecurityType = EQTY, ShareType = NS) on NYSE, AMEX, and NASDAQ. For Part B, banking stocks are identified using SIC codes 6020–6029 and 6035–6036.

Implementation

At each year-end from 1960 to 2025, a CAPM regression is estimated for each stock using the preceding five years of daily data, requiring at least 750 trading days. The systematic share equals the R^2 of the regression; the idiosyncratic share is $1 - R^2$. Shares are aggregated using lagged (prior year-end) market capitalization as weights. Regressions are evaluated at annual frequency rather than the monthly frequency likely used in the paper, which is why our plots contain fewer data points and appear less smooth; the long-run trend, however, is the same. For Part B, the broad market factor is replaced by a value-weighted banking sector excess return constructed daily from the restricted banking universe.

Part A — Broad Market

The value-weighted systematic share rises from approximately 20–25% in the early 1960s to around 35–45% in the 2010s–2020s, with the idiosyncratic share declining symmetrically. This upward trend is consistent with the central finding of Lochstoer and Muir (2025): the growth of index trading has structurally increased the share of market-level risk in individual stock returns. Our estimates are roughly 5–6 percentage points above the paper’s reported values (e.g., ~25% vs. ~19% in 1960). This discrepancy likely reflects two factors: (i) the updated CRSP field names (SecurityType/ShareType) may not map exactly onto the legacy SHRCD filter, and (ii) pre-1972 CRSP coverage is limited to NYSE stocks, which tend to be larger and more correlated with the market, mechanically raising average R^2 .

Part B — Banking Sector

The banking sector’s systematic share is substantially higher than the broad market’s throughout the sample, rising from roughly 30–40% in the late 1960s to approximately 70–77% by the 2020s. The banking series begins in the mid-1960s rather than 1960 because early CRSP coverage includes very few banking stocks, and many fail to meet the 750-day minimum observation requirement for the five-year rolling window.

The higher level relative to Part A reflects the structural homogeneity of banking: all banks are exposed to the same interest-rate environment, the same credit cycle, and a unified regulatory framework. These common factors generate strong within-sector co-movement that far exceeds what a typical stock shares with the broad market. The steeper upward trend can be attributed to several reinforcing forces operating at different times. In the 1990s, the Riegle-Neal Interstate Banking Act (1994) permitted cross-state mergers, triggering a wave of consolidation that reduced the number of U.S. commercial banks from over 14,000 to fewer than 4,200 today and homogenized surviving

banks' business models. In the 2000s, the buildup to the financial crisis saw most large banks converge on the same asset class—mortgage-backed securities—creating an endogenous portfolio similarity that became apparent when housing prices collapsed and all banks were simultaneously exposed. After 2008, post-crisis regulation (Dodd-Frank, Basel III stress tests) further standardized bank operations, while the market itself learned to price banks as a highly correlated group. The resulting systematic share of 70–77% is a direct indicator of systemic risk concentration: when one major bank is stressed, the rest tend to move in the same direction.